



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Case Study No.: 6

1. Case Study Title: To demonstrate JavaScript string operations using substring(), indexOf(), split(), replace(), string reversal, and vowel counting based on user input.

2. Case Study Program Code:

```
<!DOCTYPE html>
<html>
<head>
    <title>String Operations</title>

    <style>
        body {
            background-color: rgb(239, 171, 243);
            font-family: Arial, sans-serif;
            padding: 20px;
        }

        input {
            padding: 8px;
            margin: 5px 0 15px 0;
            width: 350px;
            display: block;
        }

        textarea {
            padding: 10px;
            margin: 5px 0 15px 0;
            width: 500px;
            height: 80px;
            display: block;
        }
```



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```
}

button {
    padding: 10px 20px;
    margin: 10px 0;
    cursor: pointer;
}

#output {
    margin-top: 20px;
    font-size: 18px;
    line-height: 1.6;
}

</style>
</head>

<body>

    <h2>Sonal's JavaScript String Operations</h2>

    <label for="str">Enter String:</label>
        <textarea id="str" placeholder="Enter your string or
paragraph"></textarea>

    <label for="start">Substring Start Index:</label>
    <input type="number" id="start" placeholder="Enter start index (e.g.,
0)">

    <label for="end">Substring End Index:</label>
    <input type="number" id="end" placeholder="Enter end index">

    <label for="search">IndexOf - Enter text to search:</label>
    <input type="text" id="search" placeholder="Enter character or word">
```



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```
<label for="separator">Split - Enter separator:</label>
  <input type="text" id="separator" placeholder="Type a character or
'space'">

<label for="oldText">Replace - Text to replace:</label>
<input type="text" id="oldText" placeholder="Enter text to replace">

<label for="newText">Replace - New text:</label>
<input type="text" id="newText" placeholder="Enter new text">

<button onclick="performOperations()">Perform Operations</button>

<div id="output"></div>

<script>
function performOperations() {
  let str = document.getElementById("str").value;

  if (!str) {
    document.getElementById("output").innerHTML = "<strong>Please
enter a valid string first!</strong>";
    return;
  }

  let start = Number(document.getElementById("start").value);
  let end = Number(document.getElementById("end").value);

  let search = document.getElementById("search").value;
  let separator = document.getElementById("separator").value;

  let oldText = document.getElementById("oldText").value;
  let newText = document.getElementById("newText").value;

  let output = "";
```



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```
// Substring (Handles cases where end index isn't specified)
    let subResult = (end > 0) ? str.substring(start, end) :
str.substring(start);
    output += "<strong>Substring:</strong> " + subResult + "<br>";

// IndexOf
if (search) {
    output += "<strong>Index of '" + search + "':</strong> " +
str.indexOf(search) + "<br>";
}

// Split
if (separator) {
    let sepChar = (separator === "space") ? " " : separator;
    output += "<strong>Split:</strong> [" + str.split(sepChar).join(",
") + "]"<br>";
}

// Replace (Replaces all instances safely)
if (oldText) {
    output += "<strong>Replace:</strong> " + str.replaceAll(oldText,
newText) + "<br>";
}

// Reverse
let reverse = str.split("").reverse().join("");
output += "<strong>Reversed String:</strong> " + reverse + "<br>";

// Count Vowels (Using regex for cleaner execution)
let vowelMatch = str.match(/[aeiou]/gi);
let count = vowelMatch ? vowelMatch.length : 0;

output += "<strong>Number of Vowels:</strong> " + count;
```



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```
document.getElementById("output").innerHTML = output;
}
</script>
</body>
</html>
```

3. Output:

Sonal's JavaScript String Operations

Enter String:

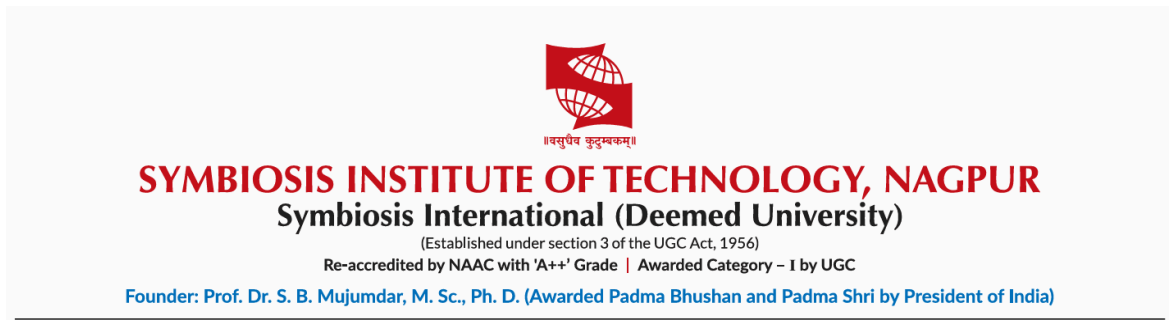
Substring Start Index:

Substring End Index:

IndexOf - Enter text to search:

Split - Enter separator:

Replace - Text to replace:

The image is a screenshot of a web browser displaying a JavaScript application for string operations. The browser's address bar shows the file path 'C:/Users/Sonal/Desktop/Documents/JS_24070521199/Exp6/CS.html'. The application has a light purple background. On the left side, there are four input sections: 'IndexOf - Enter text to search:' with a text box containing 'Sonal'; 'Split - Enter separator:' with an empty text box; 'Replace - Text to replace:' with a text box containing 'my name is'; and 'Replace - New text:' with a text box containing 'I am'. Below these is a 'Perform Operations' button. On the right side, the results of the operations are displayed: 'Substring: , my', 'Index of 'Sonal': 18', 'Split: [Hello,, my, name, is, Sonal, Nakhate.]', 'Replace: Hello, I am Sonal Nakhate.', 'Reversed String: .etahkaN lanoS si eman ym ,olleH', and 'Number of Vowels: 10'.

4. Result/Conclusion: Thus, various JavaScript string operations were successfully performed, including substring extraction, text searching, splitting, replacing, reversing, and vowel counting, and the results were displayed dynamically on the webpage.